ShivShreeram P

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	8.45
Yak Public School, Khopoli	Class XII (CBSE)	2023	93.00%
Atomic Energy Central School #4, Mumbai	Class X (CBSE)	2021	93.00%

EXPERIENCE

Technical Intern, Autoregressive Image Diffusion (AID)

Siemens Healthineers, Bengaluru

- **Improved AID MRI reconstruction** at 6x acceleration over the original paper's results: **+1.9% PSNR, +3% SSIM, and an FID of 36**, with further gains at 8x.
- **Identified and fixed a critical bug** in the data-consistency pipeline that was silently degrading reconstruction quality and had passed standard checks.
- **Benchmarked standard vs. custom data-consistency algorithms** and ran **ablation studies** to improve diffusion model stability and inference performance.
- **Leading the ongoing effort** to extend AID's supported acceleration range from 2x/4x to 6x, optimizing the diffusion algorithms and inference pipeline.

Analytical Modeling of 3D Volumetric Data

Research Intern - Under Dr.Gaurav Bhutani, BARC, Mumbai

- Contributed to **3D representation of MRI scans** using volume rendering, calibrated to real-world coordinates for precise biomedical spatial analysis.
- Applied mathematical and algorithmic techniques to identify optimal voxel intensity ranges representing anatomical structures.
- Developed efficient segmentation and volumetric processing pipelines, reducing computational load for large biomedical datasets.
- Enabled **interactive point-picking and visualization**, supporting morphological inspection similar to microscopy-style analysis.
- Followed iterative improvement principles (PDCA cycle) to refine processing workflows under research supervision.

StrainNet: Soft Tissue Deformation Analysis

Research Intern -Under Prof. Sai Sidhardh, IIT, Hyderabad

- Contributed to StrainNet, a deep learning framework modeling **soft-tissue deformation** using controlled synthetic biomedical datasets.
- Implemented **blob detection** to track particle-like elements and understand deformation behavior, analogous to tracking structures in microscopy images.
- Built CNN-based DeformationNet and U-Net systems achieving **98% training and 85% validation accuracy** across deformation classes.
- Designed data preprocessing, cleaning, and visualization steps in **PyTorch/OpenCV** to analyze strain maps and deformation fields.
- Collaborated with mentors to refine models, document findings, and integrate improvements into iterative training cycles.

PROJECTS

Translation & Language Automation

- Built an **end-to-end NLP pipeline** using Hugging Face Transformers for multilingual language detection and translation (100+ languages).
- Applied **XLM-RoBERTa and M2M100** models for high-accuracy (93%) text translation.
- Demonstrated potential use for **automating multilingual regulatory submissions and harmonized document handling**.

MRI Brain Tumor Classification using CNNs

- Developed a **CNN-based classifier** with **97.6% accuracy** on MRI brain scan datasets.
- Led complete ML pipeline: data preprocessing, augmentation, model building with **Keras/TensorFlow**, and evaluation.
- Applied **regularization techniques** like **dropout** and **batch normalization** to improve generalization.

SKILLS

GenAI & LLMs: LangChain, RAG Pipelines, LLM Agents, Hugging Face Transformers

Programming & Scripting: Python, C, C++, R, Verilog, HTML/CSS

AI & Medical Imaging: Deep Learning, Image Processing, DICOM File Management, VTK Visualization

Frameworks & Tools: TensorFlow, Keras, PyTorch, OpenCV, FAISS/Chroma, FastAPI, CAD Modeling, SPICE/TINA Simulation

Graphics & Simulation: Computer Graphics, Raycasting, Marching Cubes, Monte Carlo Simulation

Leadership & Communication: Team Management, Event Execution, Resource Planning

ACHIEVEMENTS

Smart India Hackathon - 2024: Selected as one of **4 teams nominated** by the institute; developed a **non-contact device for glaucoma detection** using **photoelasticity-based eye deformation analysis**.

Mercedes-Benz Safe Roads Ideathon - 2024: Ranked among **Top 10 college teams**; designed **real-time data-driven alert models** using weather, sensor, and map data, showcasing **predictive modeling and automation skills**.

RELEVANT COURSES

Artificial Intelligence (8/10), **Data Structures and Algorithms Lab** (8/10),**Introduction to Biomedical Imaging** (9/10)

Analog and Integrated Circuits (10/10), **Introduction to Embedded Systems** (8/10), **Bioinformatics** (9/10)